

SHIVANK SAPRA

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EDUCATION

University of Minnesota - Twin Cities

May 2027 (Expected)

Master of Science, Computer Science (4.0 GPA)

Coursework: Machine Learning, Natural Language Processing, Computer Vision, Web Development, Computational Genomics

Guru Gobind Singh Indraprastha University, New Delhi

July 2023

Bachelor of Technology, Computer Science & Engineering

SKILLS

Languages: Python, C++, JavaScript, C#, SQL

AI & Machine Learning: Computer Vision, Machine Learning, Deep Learning, Reinforcement Learning

Frameworks & Libraries: TensorFlow, NumPy, Pandas, OpenCV, Matplotlib

Tools & Web Frameworks: Flask, Vue.js, Django, .NET, MVC Architecture, SQL, PostgreSQL, Firebase, Git & Github

PROJECTS

Real-Time Scoring for Rhubarb Dice Game [{Link}](#) | OpenCV, YOLOv8, Computer Vision

- Built a real-time computer vision pipeline to detect & track cards & dice under rotation, occlusion, & motion blur using YOLOv8
- Improved small-object detection by separating detection, dice cropping, color classification, and value classification
- Implemented multi-object tracking (IoU, Kalman, EMA) and stabilized scoring using per-frame state recomputation; evaluated with mAP and precision/recall.

Adaptive Traffic Congestion Control [{Link}](#) | Python, Reinforcement Learning

- Designed and trained a Deep Q-Learning model to dynamically adjust signal timings at a 4-way intersection based on real-time traffic.
- Built an end-to-end ML pipeline for state ingestion, policy learning, and performance evaluation.
- Applied neural networks to learn optimal action selection under uncertainty.

Image Super-Resolution [{Link}](#) | Python, Tensorflow, GANs

- Implemented and trained a Super-Resolution GAN (SRGAN) for 4× image enhancement.
- Designed perceptual loss functions using features from pre-trained deep neural networks to create photo-realistic images with enhanced texture and detail.

ClashPoint - Online Debating Forum [{Link}](#) | Flask, PostgreSQL, Tensorflow.js

- Developed the Clash Page for a Flask-based application, enabling users to view, post, reply, edit, and delete arguments in real time.
- Integrated real-time machine learning based toxicity detection (Tensorflow.js) to improve user safety and engagement.
- Built scalable backend logic for threaded discussions (recursive AJAX), voting, and ML-driven moderation workflows.

EXPERIENCE

Software Development Engineer | Teleperformance

November 2023 - May 2025

- Developed and maintained scalable backend systems and APIs for data-driven applications. (C#, .NET Framework, MVC architecture, HTML, CSS, JavaScript)
- Optimized SQL Server stored procedures and indexing strategies, reducing query latency by ~50%.
- Led the development of a collaborative real-time document editor (ReactJS, TypeScript, Express, Socket.io, PostgreSQL), ensuring seamless synchronization.

Summer Intern | GGSIPU USS ACM

July 2022 - September 2022

- Developed a decentralized system to mint academic credentials as NFTs on the Ethereum blockchain securely.
- Designed web interface (MERN stack, Web3.js) and smart contracts using Solidity to manage document uploads, NFT minting, and wallet-based certificate transfers with guaranteed transparency.

Software Development Engineer Intern | Teleperformance

December 2021 - May 2022

- Developed a Computer Vision-based Attendance System using Django and OpenCV for seamless attendance management.
- Designed secure role-based access control (Admin/User) to enable personalized dashboards and protected data access.

Machine Learning Intern | GGSIPU USS ACM

August 2021 - September 2021

- Developed a Real-time Drowsiness Detection System with face and eye detection using OpenCV and Haar Cascade Classifiers to enhance road safety by alerting drivers showing signs of fatigue.
- Designed and trained a stacked deep convolutional neural network (CNN) and employed MobileNet for transfer learning to extract dynamic features from keyframes in video sequences. Integrated a Sigmoid classifier layer in the CNN architecture to predict the driver's state as drowsy or alert, triggering an alarm when drowsiness was detected continuously.